

Programming Reference: C Language Fundamentals

Educational Series

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1 Introduction to C

C is a general-purpose, procedural computer programming language developed in 1972 by Dennis Ritchie at Bell Labs. It is widely used for system programming, developing operating systems, and embedded systems because of its efficiency and low-level memory access.

2 Expanded Data Types

C has a rich set of data types. Beyond the basics, we use **Type Modifiers** to alter the meaning of base types:

- **Primary Types:** `int` (Integer), `char` (Character), `float` (Floating point), `double` (Double precision).
- **Modifiers:** `signed`, `unsigned`, `short`, `long`.
- **Example:** An `unsigned int` can only store positive values, effectively doubling the positive range compared to a `signed int`.

3 Functions

Functions are self-contained blocks of code that perform a specific task. They help in code reusability and modularity.

- **Declaration:** Tells the compiler about the function name, return type, and parameters.
- **Definition:** The actual body of the function.

```
1 int add(int a, int b) { // Return type is int
2     return a + b;
3 }
4
5 int main() {
6     int sum = add(10, 20); // Function call
7     return 0;
8 }
```

Listing 1: Function Example

4 Arrays

An array is a collection of data items of the **same type** stored at contiguous memory locations.

- **Index:** Starts from 0. An array of size N has indices from 0 to $N - 1$.
- **Declaration:** `int marks[5];` (Creates an array for 5 integers).

```
1 int numbers[] = {10, 20, 30, 40};
2 int first = numbers[0]; // Accessing the first element
```

5 Strings

In C, a string is simply an **array of characters** terminated by a null character `'\0'`.

- **Declaration:** `char name[] = "Hello";`
- **Library:** `string.h` provides functions like `strlen()` (length), `strcpy()` (copy), and `strcmp()` (compare).

6 Control Flow Summary

- **if...else:** Conditional branching.
- **switch:** Multi-way branching based on constant values.
- **for / while / do-while:** Iterative loops for repetitive tasks.

7 Standard I/O Example

```
1 #include <stdio.h>
2 #include <string.h>
3
4 int main() {
5     char greeting[20] = "Hello";
6     int age;
7
8     printf("Enter your age: ");
9     scanf("%d", &age);
10
11     if (age >= 18) {
12         printf("%s, you are an adult.\n", greeting);
13     } else {
14         printf("%s, you are a minor.\n", greeting);
15     }
16
17     return 0;
18 }
```

Listing 2: Comprehensive C Example